ME72 Engineering Design Laboratory

Design Competition

Dredge Challenge

Competition Rules

Department of Mechanical and Civil Engineering

California Institute of Technology

Fall 2026 – Winter 2027

Revision 1

1 Competition Overview and Design Objectives

The 2026–27 ME72 Design Competition, *Dredge Challenge*, will be played within the approximately 42-ft by 51-ft section of Caltech Hall Pond adjacent to the east face of Caltech Hall, as shown in Fig. 1. Each team will design and construct three robotic systems to collect, transport, and recover golf balls distributed both at the water surface and on the bottom of the pond. Two robots will operate within the pond to collect and transport golf balls to designated staging areas, while a third, dedicated transfer robot will complete the recovery process by moving the collected golf balls from these staging areas to a dry-dock scoring area on the adjacent concrete walkway.



Figure 1: **Caltech Hall Pond and the ME72 competition field.** The *Dredge Challenge* will take place within an approximately 42-ft by 51-ft region of Caltech Hall Pond adjacent to the east face of Caltech Hall. (a) View from the southeast. (b) Aerial view (north up). The 42-ft dimension extends from the easternmost face of the octagonal building to the eastern edge of the pond. Dimensions are approximate.

The active playing field will be bounded to the north and south by the existing concrete pond ledges. Floating barriers will define the eastern and western limits of play and prevent floating golf balls from leaving the competition area. The eastern floating barrier will be aligned approximately with the pedestrian bridge, while the western floating barrier will be aligned with the east-facing edge of the octagonal foundation of Caltech Hall. Detailed barrier locations and field geometry are described in Section 2.

Two classes of golf balls will be used:

- **Submerged golf balls**, consisting of conventional golf balls that sink and will be distributed across the textured pebble surface of the pond bottom; and
- **Floating golf balls**, consisting of purpose-designed floating golf balls having the same nominal diameter as a conventional golf ball but a reduced mass that allows them to remain buoyant at the water surface.

The two classes of golf balls will be distinguished by color. The floating golf balls will be white, while the submerged golf balls will be orange, reflecting Caltech's traditional colors. This distinction will facilitate identification during competition and scoring.

The floating competition balls will be HydroStar floating golf balls manufactured by EverySwing Golf. These balls have a nominal diameter of 1.680 in. (42.67 mm), a mass of 40 g (1.41 oz), and a two-piece Surlyn construction. A lightweight floating core reduces the overall density of the ball sufficiently for it to remain buoyant at the water surface while retaining the dimensions and general handling characteristics of a conventional golf ball.

The competition is intended to emphasize the design and integration of mechanical systems operating in a shallow-water environment and across the interface between the pond and the surrounding concrete terrain. Relevant engineering considerations include flotation and stability, propulsion, waterproofing, object acquisition, dredging, storage, unloading, material transfer, mechanism design, navigation of the water–land transition, and overall system integration.

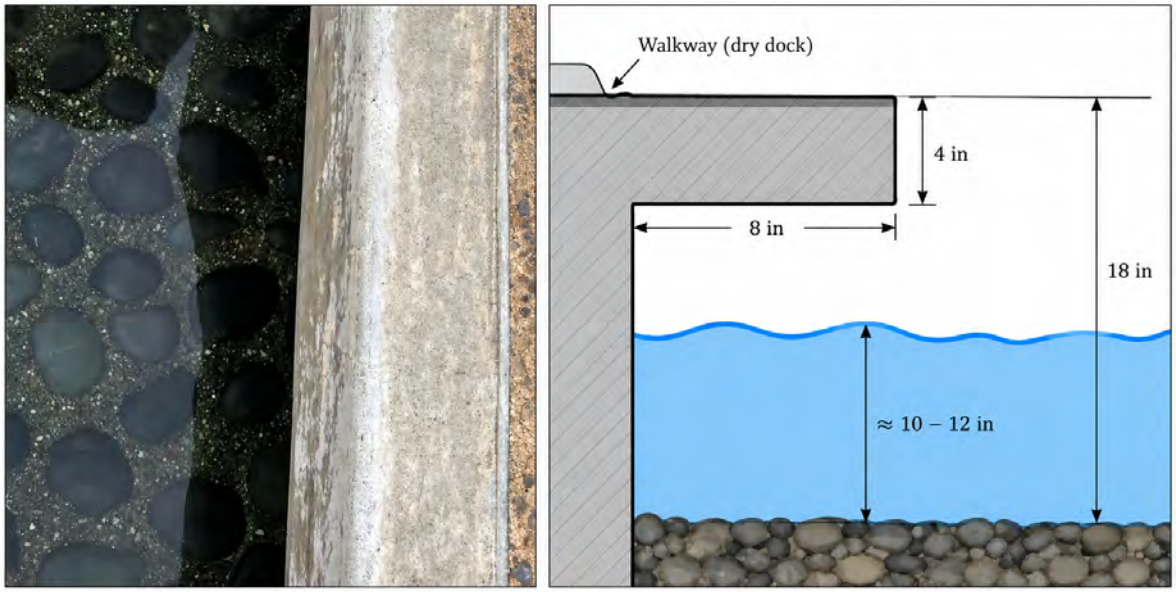


Figure 2: **Caltech Hall Pond ledge geometry.** (a) Photograph of the concrete pond ledge and the textured pebble surface of the pond bottom. (b) Schematic cross-section illustrating the geometry of the concrete walkway and its overhanging ledge. The walkway surface is approximately 18 in. above the pond bottom, and the concrete ledge projects approximately 8 in. horizontally over the pond. The ledge is approximately 4 in. thick, while the nominal water depth is approximately 10–12 in. Dimensions are approximate.

Figure 2 illustrates the approximate ledge geometry and representative pond-bottom conditions. The concrete walkway surrounding Caltech Hall Pond is elevated above the water surface and forms a distinct ledge around the perimeter of the playing field. This ledge separates the in-water recovery staging areas from the dry-dock scoring area and represents

an important geometric constraint on the final ball-transfer task.

Each team must develop three robotic systems: two collection bots and one dedicated transfer bot. The collection bots will recover golf balls from the playing field and deliver them to the team's designated recovery staging areas. The transfer bot will complete the recovery process by moving golf balls from these staging areas to the designated dry-dock scoring region on the adjacent concrete walkway. The transfer bot may be stationary or mobile, subject to the operating restrictions specified in Section 3.

The two collection bots may be specialized for different tasks, such as floating-ball collection and submerged-ball dredging, or may be designed with overlapping capabilities. Teams are free to determine how these collection functions are distributed between their two bots.

Each team will be assigned a designated harbor at its north or south end of the playing field. The harbor will serve as a protected region for launching and recovering collection bots and for delivering recovered golf balls to the team's staging areas. A harbor-divider barrier will separate each harbor from the central playing region, with a 4-ft-wide entrance aligned with the team's starting zone.

Within each harbor, one staging area will serve as the floating-ball containment area, while the other will serve as the submerged-ball receiving platform. These areas provide designated locations for delivering recovered golf balls and serve as intermediate scoring regions. Their geometry and construction are described in Section 2.

Golf balls delivered to the appropriate staging areas may earn intermediate-recovery points. Additional scoring credit is available when the transfer bot successfully moves recovered golf balls onto the adjacent concrete containment area, with the highest scoring values awarded to balls deposited within the designated dry-dock scoring pen. The collection bots may not transfer golf balls directly onto the concrete walkway. Scoring is based on the final location of each golf ball at the end of the heat.

Because recovering submerged golf balls introduces additional mechanical and operational difficulty, these balls carry greater scoring value than floating golf balls. The competition's scoring structure rewards both successful collection and completion of the final pond-to-land transfer task. Detailed scoring provisions are presented in Section 5.

2 Competition Field and Game Elements

2.1 Playing Field Geometry

The active competition field will occupy an approximately 42-ft by 51-ft rectangular region of Caltech Hall Pond, as introduced in Fig. 1. The 42-ft dimension extends approximately east–west, while the 51-ft dimension extends approximately north–south. Teams will operate from opposite ends of the field, with one team assigned to the north end and the other to the south end during each heat.

The pond is shallow, with a nominal water depth of approximately 10–12 in. across the active

playing region. The pond bottom consists of a textured surface formed by rounded pebbles of varying sizes, shapes, and spacing, with some local variation in depth. Representative bottom conditions are illustrated in Fig. 3.

Unlike a smooth, flat bottom, the pebble surface provides an uneven foundation on which submerged golf balls will come to rest. Some balls may settle into depressions between adjacent pebbles, while others may remain partially elevated on the surrounding stones. The local height and support conditions of a submerged golf ball may therefore vary considerably across the playing field.

These conditions present an important challenge for the design of submerged-ball acquisition mechanisms. A scoop or intake positioned too high above the pond bottom may pass over balls nestled between the pebbles, while a mechanism operating too close to the surface may encounter interference, increased friction, or jamming. Successful designs may need to accommodate these variations through suitable intake geometry, compliant elements, adjustable positioning, or other mechanical features.



Figure 3: **Textured pebble surface of Caltech Hall Pond.** (a) Close-up view of the irregular pebble-covered bottom. (b) View of the pebble surface adjacent to the concrete pond ledge.

The northern and southern boundaries of the playing field are defined by the existing concrete pond ledges. The eastern and western boundaries will be established using floating barriers to prevent floating golf balls from leaving the active playing region. The proposed floating barriers will consist of a series of cylindrical sections of closed-cell foam, such as expanded polystyrene, connected by a rope or cable passing through their longitudinal axes. The eastern barrier will be aligned approximately with the pedestrian bridge, while the western barrier will be aligned with the east-facing edge of the octagonal foundation of Caltech

Hall. These barriers will be secured to maintain their intended positions throughout the competition. Representative visualizations of the proposed eastern and western floating barriers are shown in Fig. 4.



Figure 4: **Proposed floating boundaries of the competition field.** Photographic visualizations illustrating the proposed placement of floating barriers at Caltech Hall Pond. The western boundary is aligned with the east-facing edge of Caltech Hall's octagonal foundation, while the eastern boundary is aligned approximately with the pedestrian bridge. The barriers define the east–west limits of the active playing field and prevent floating golf balls from leaving the competition area.

The competition-field layout is illustrated in Fig. 5. Identical team regions are located at the north and south ends of the field.

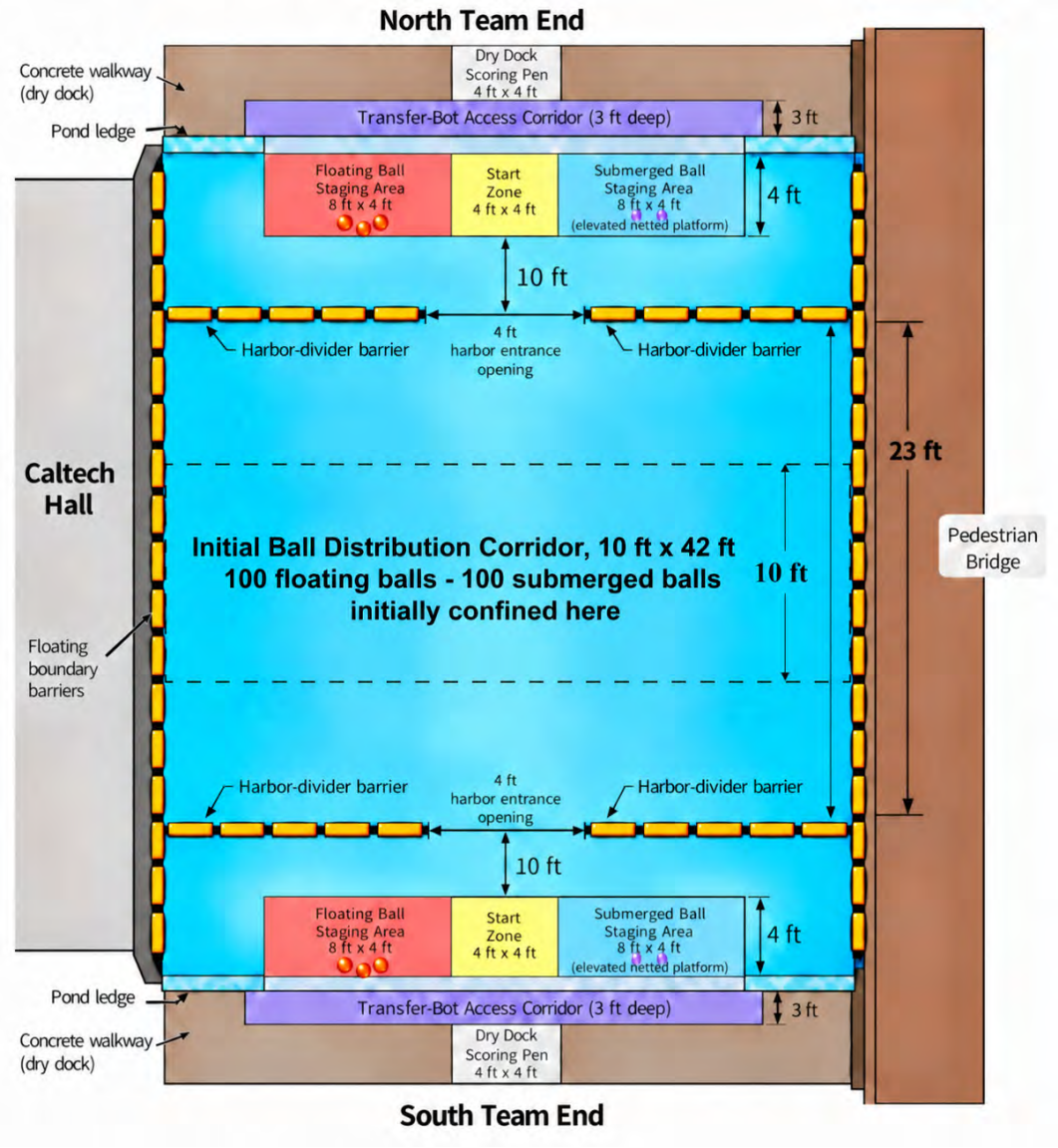


Figure 5: **Competition field layout.** Plan view of the approximately 42-ft by 51-ft active playing field showing the north and south team harbors, each separated from the central playing region by a harbor-divider barrier with a 4-ft-wide entrance. Each harbor includes a central 4-ft by 4-ft starting zone, an 8-ft by 4-ft floating-ball staging area to the west, and an 8-ft by 4-ft submerged-ball staging area to the east. The central 42-ft by 10-ft initial ball-distribution corridor is bounded by temporary floating barriers that are removed at the start of each heat. Adjacent concrete walkways provide transfer-bot access and dry-dock scoring areas. Dimensions are approximate.

Two team harbors will be established at the north and south ends of the playing field. Each harbor will be separated from the central playing region by a transverse harbor-divider barrier extending east–west across the pond, except for a central 4-ft-wide entrance. The entrance will be aligned with the midpoint of the 42-ft field width, approximately 21 ft from either eastern or western boundary.

Each harbor-divider barrier will be positioned approximately 10 ft outward from the front edge of its associated staging and starting zones. Since these zones extend 4 ft into the pond, the resulting harbor will have an overall north–south depth of approximately 14 ft. The two harbor-divider barriers will therefore be separated by approximately 23 ft, forming the central playing region.

The initial golf-ball distribution corridor will occupy an approximately 42-ft by 10-ft strip centered within the 23-ft central playing region, extending across the east–west width of the pond. Both floating and submerged golf balls will initially be concentrated within this region. Temporary floating barriers will confine the floating balls before the start of each heat and will be removed by course staff at the starting signal. Detailed starting conditions and ball quantities are specified in Section 4.

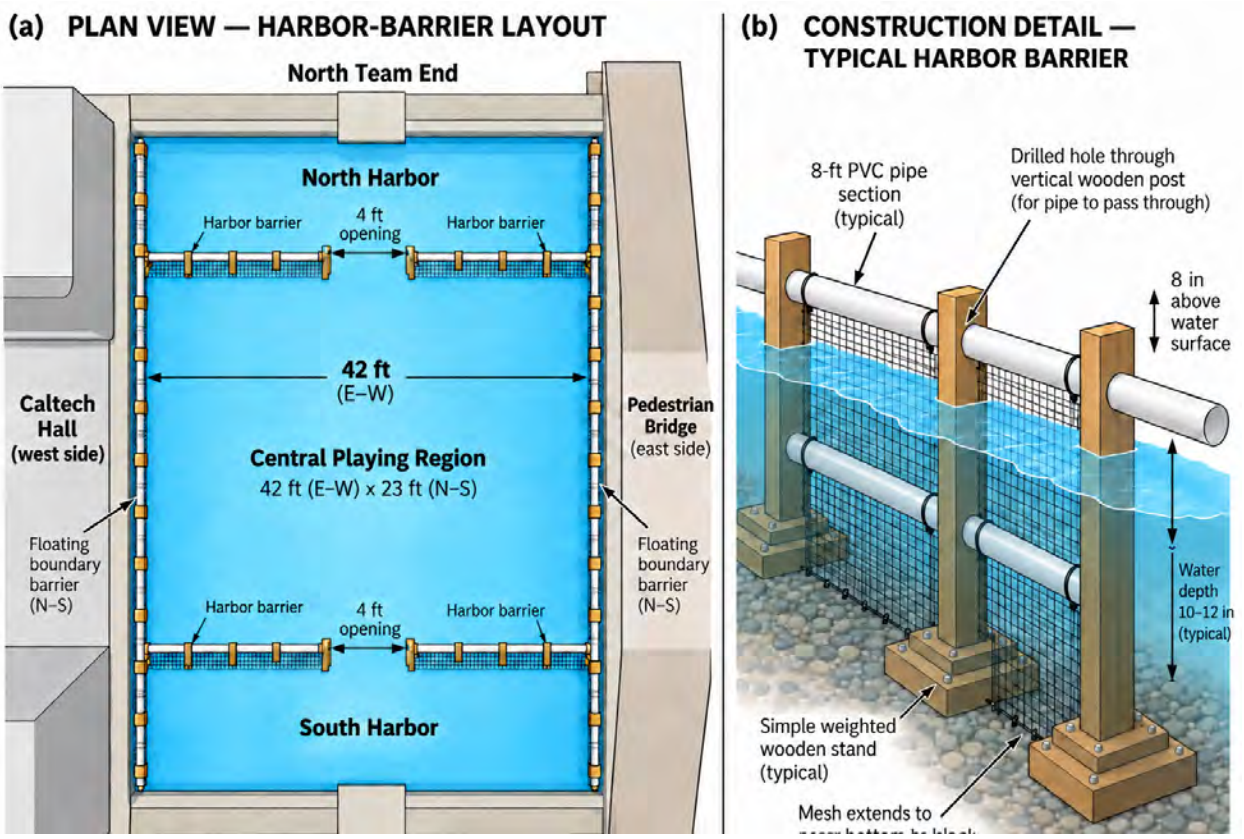


Figure 6: **Harbor-divider barrier layout and representative construction.** (a) Plan view showing the north and south team harbors, each separated from the central playing region by a transverse barrier with a 4-ft-wide entrance. (b) Construction detail illustrating horizontal PVC pipes passing through drilled vertical wooden supports, with wire mesh extending from above the water surface to the pond bottom. The uppermost PVC member is approximately 8 in. above the nominal water surface. Weighted support bases provide stability. Dimensions and construction details are approximate.

The harbor-divider barriers are intended to limit the migration of floating and submerged golf balls into the team harbors while providing a defined entrance through which collection

bots may depart and return. Each barrier will span the 42-ft east–west width of the pond, interrupted by a 4-ft-wide opening at the field centerline. The two barrier sections on either side of the opening will therefore have nominal lengths of 19 ft each.

The proposed construction will consist of horizontal PVC pipes supported by vertical wooden posts. The PVC pipes will pass through drilled holes in the posts, forming a simple framework with multiple horizontal members. The pipes may be assembled from standard 8-ft lengths using appropriate connectors.

Wire mesh, such as chicken wire or galvanized hardware cloth, will be attached to the framework to prevent golf balls from passing through the barrier. The uppermost PVC member will extend approximately 8 in. above the nominal water surface, while the mesh will extend downward to the pebble-covered pond bottom. This arrangement is intended to limit the migration of both floating and submerged golf balls into the team harbors.

The vertical posts will be supported by weighted bases resting on the pond bottom. The framework will be designed to maintain its intended position and withstand incidental contact with collection bots without collapsing or becoming displaced. Representative geometry and construction are illustrated in Fig. 6.

2.2 Golf Balls

Two classes of golf balls will serve as the competition game pieces. A total of 200 golf balls will be used at the start of each heat, consisting of 100 submerged golf balls and 100 floating golf balls.

- **Submerged golf balls** will consist of orange, conventional-weight golf balls having a nominal diameter of 1.680 in. (42.67 mm) and an advertised mass of approximately 45 g. These balls will sink and rest on the textured pebble surface of the pond bottom.
- **Floating golf balls** will consist of white HydroStar floating golf balls manufactured by EverySwing Golf. These balls have a nominal diameter of 1.680 in. (42.67 mm), an advertised mass of approximately 40 g (1.41 oz), and a two-piece Surlyn construction. Their reduced average density allows them to remain buoyant at the water surface.

The orange and white colors reflect Caltech's traditional colors and provide a clear visual distinction between the two classes of golf balls during competition and scoring.

At the beginning of each heat, the golf balls will be concentrated within an approximately 42-ft by 10-ft rectangular distribution region near the center of the playing field, as shown in Fig. 5. The floating golf balls will initially be confined at the water surface by temporary removable barriers, while the submerged golf balls will be distributed on the pond bottom within the same region.

The temporary barriers will be removed by course staff at the start of each heat. Additional starting procedures are described in Section 4.

2.3 Ball Recovery Staging Areas

Each team will have two designated ball recovery staging areas and one starting zone within its assigned harbor. The arrangement of these areas will be identical at the north and south ends of the playing field, as illustrated in Fig. 7.

The 42-ft-wide north and south boundaries have an east–west midpoint located 21 ft from either side of the playing field. A 4-ft by 4-ft starting zone will be centered on this midpoint. Two recovery staging areas will be located immediately adjacent to the starting zone, one to the east and one to the west.

Each staging area will have nominal plan dimensions of 8 ft by 4 ft. The 8-ft dimension extends east–west along the team boundary, while the 4-ft dimension extends into the pond. The western staging area will serve as the floating-ball containment area, while the eastern staging area will serve as the submerged-ball receiving area.

The staging areas constitute the intermediate recovery regions of the competition. Golf balls delivered to their appropriate staging areas may earn scoring credit even if they are not subsequently transferred to the dry dock.

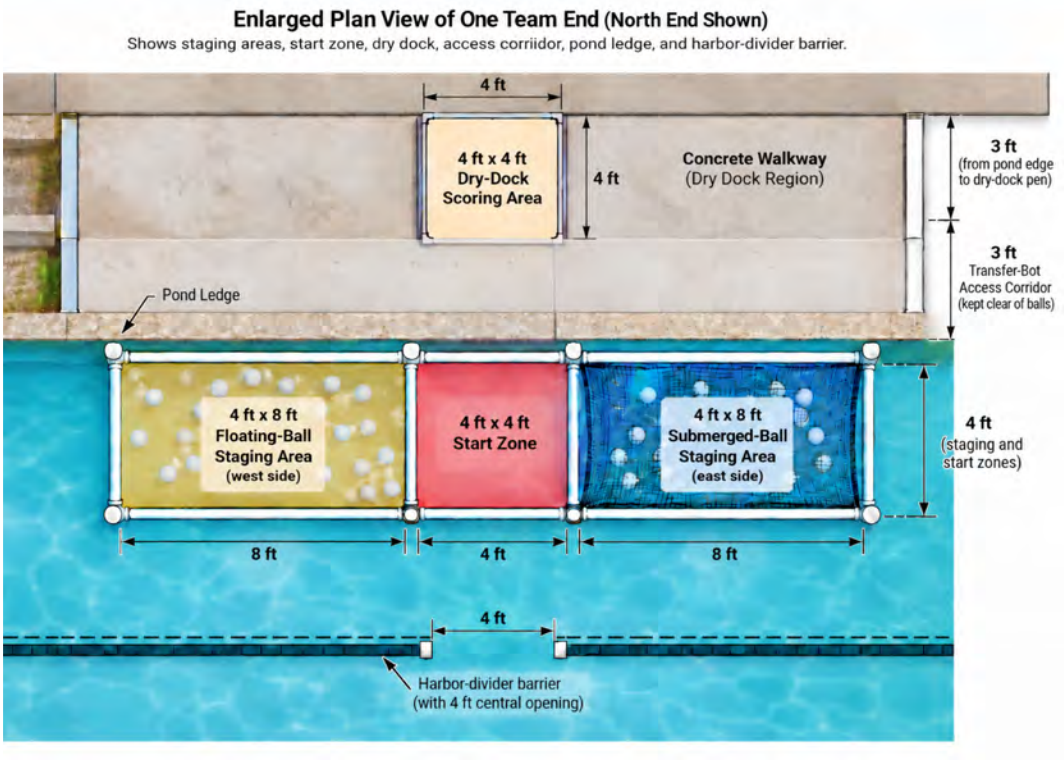


Figure 7: **Team harbor, staging areas, and starting-zone geometry.** Enlarged plan view of the north team harbor showing the central 4-ft by 4-ft starting zone, the adjacent 8-ft by 4-ft floating-ball and submerged-ball staging areas, and the harbor-divider barrier with its centrally aligned 4-ft-wide entrance. Also shown are the concrete pond ledge, transfer-bot access corridor, and 4-ft by 4-ft dry-dock scoring pen. The south team harbor will have an equivalent arrangement. Dimensions are approximate.

Floating-Ball Containment Area

The floating-ball staging area will consist of an 8-ft by 4-ft open-water enclosure. Its perimeter will be defined by a removable, semi-rigid containment structure consisting of horizontal PVC rails supported by vertical posts with weighted or anchored bases resting on the pond bottom.

Lightweight mesh or netting will be installed between the horizontal rails to prevent floating golf balls from escaping through gaps in the enclosure. The uppermost PVC rail will be positioned approximately 2–3 in. above the nominal water surface to minimize the height that collection bots must overcome when depositing golf balls into the enclosure.

Floating golf balls will remain at the water surface within the enclosure. A floating golf ball that enters and remains within the designated containment area at the conclusion of a heat will receive intermediate recovery credit, regardless of whether it arrived through deliberate collection, wave action, or incidental contact.

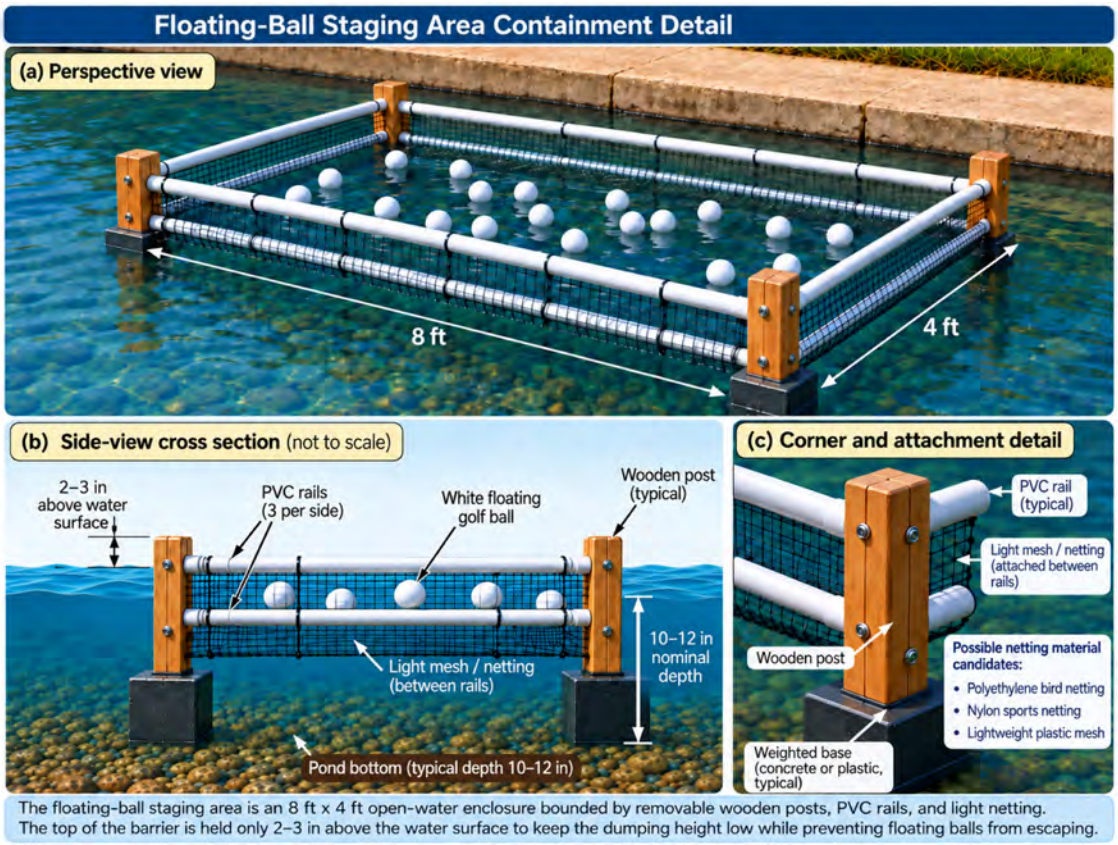


Figure 8: **Floating-ball containment area.** Representative construction of the 8-ft by 4-ft open-water enclosure. PVC rails, vertical supports, and lightweight mesh retain floating golf balls while maintaining a low receiving height above the water surface. The uppermost rail is positioned approximately 2–3 in. above the nominal waterline. Construction details are illustrative.

Submerged-Ball Receiving Net

The submerged-ball staging area will consist of an 8-ft by 4-ft receiving net suspended from a rigid rectangular perimeter frame. The receiving surface will be constructed from commercially available golf netting with mesh openings sufficiently small to retain conventional golf balls.

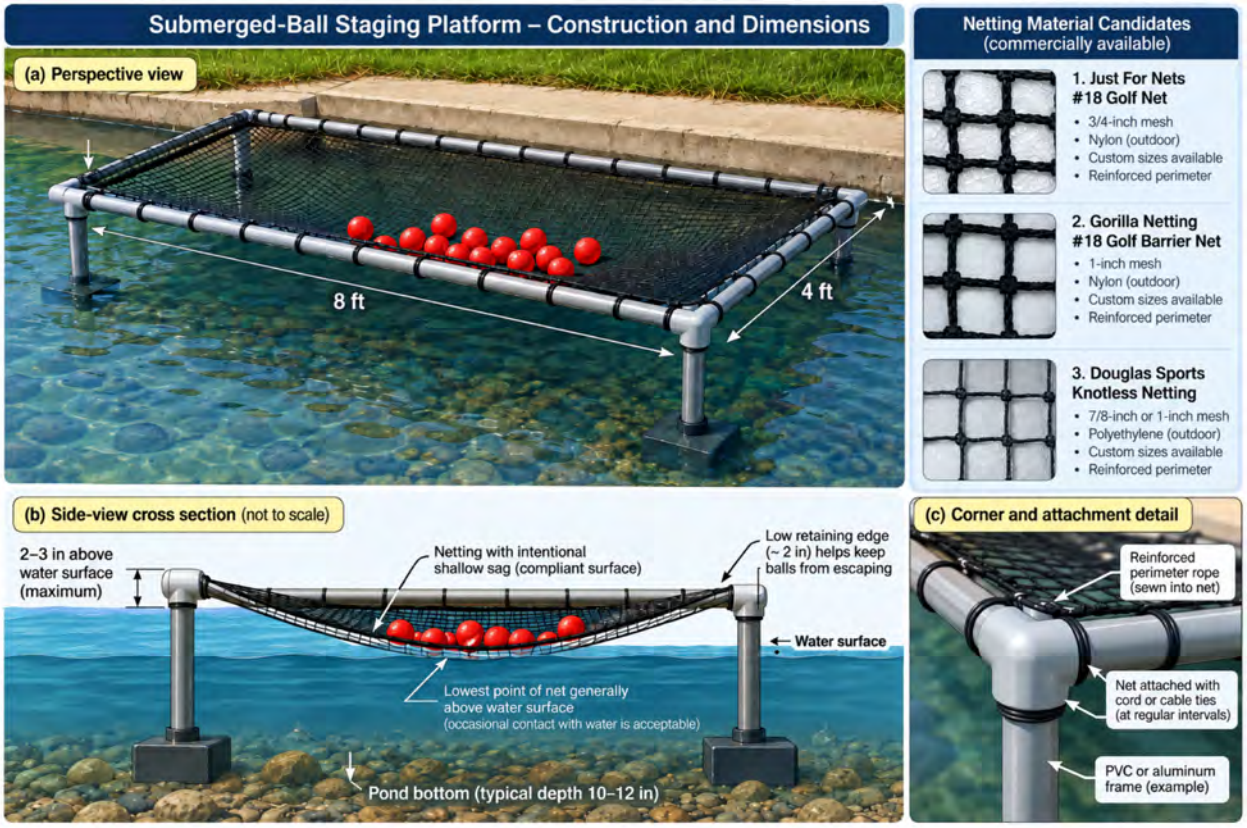


Figure 9: **Submerged-ball receiving net.** Representative 8-ft by 4-ft receiving structure consisting of a rigid perimeter frame supporting intentionally slack golf netting. The compliant net forms a shallow depression that retains recovered submerged golf balls. The frame is positioned approximately 2–3 in. above the nominal waterline, while the net may sag nearly to the water surface. Materials and construction details are illustrative.

The netting will be installed with intentional slack, forming a shallow, compliant depression into which recovered submerged golf balls can be deposited. This arrangement will allow the net to yield under the weight and impact of incoming golf balls rather than behave as a taut receiving surface.

The supporting frame will be positioned approximately 2–3 in. above the nominal water surface. The net will sag below the frame and generally remain above the water, although occasional contact with the water surface is acceptable. A low retaining edge may be incorporated around the perimeter to reduce the likelihood of deposited balls escaping.

The receiving frame will be supported by removable posts or other suitable structural members, with weighted or anchored bases resting on the pond bottom.

Submerged golf balls will not receive intermediate recovery credit merely because they have been displaced into the general vicinity of the staging area. To qualify for intermediate recovery credit, a submerged golf ball must remain within the designated receiving net at the conclusion of the heat. Golf balls resting on the pond bottom beneath or adjacent to the net will not receive such credit.

Delivery and Transfer

The collection bots are responsible for delivering recovered golf balls to their team's designated staging areas. The transfer bot is responsible for completing the final pond-to-land transfer to the designated dry-dock scoring region.

The main body of each collection bot, including its hull or flotation structure, must remain outside the designated floating-ball containment enclosure. However, both collection bots and the transfer bot may extend their ball-collection, handling, and transfer mechanisms into the designated staging areas as necessary.

The collection bots may not transfer golf balls directly onto the adjacent concrete walkway. Detailed operating restrictions are specified in Section 3.

All golf balls will be scored according to their final location or qualifying possession status at the conclusion of each heat, as specified in Section 5.

2.4 Starting Locations

At the beginning of each heat, each team's two collection bots must be positioned within the designated 4-ft by 4-ft starting zone at the team's assigned north or south end of the playing field.

The starting zone will be centered on the midpoint of the 42-ft team boundary and will extend 4 ft into the pond. Both collection bots must fit entirely within this zone at the start of the heat. Their initial orientations and relative positions may be selected by the team, subject to the dimensional restrictions specified in Section 3.

At the start of each heat, collection bots will depart their assigned harbor through the designated 4-ft-wide entrance in the harbor-divider barrier. Returning collection bots will use the same entrance to reach their staging areas. Teams must account for this opening when selecting robot dimensions, deployment mechanisms, and maneuvering strategies.

The dedicated transfer bot may begin on the concrete walkway within the team's designated operating region. Alternatively, it may be positioned in the water along the rear portion of the team's starting zone, adjacent to the pond ledge. A transfer bot positioned in the water must remain stationary throughout the heat.

The precise starting procedures and conditions are described in Section 4.2.

2.5 Dry Dock and Concrete Containment Area

Each team will be assigned a designated concrete containment area along the north or south end of the playing field, immediately adjacent to its ball recovery staging areas. This region will provide the operating surface for a land-based transfer bot and will contain golf balls that have been transferred from the pond onto the concrete walkway.

Outer Containment Area

The designated concrete containment area will have nominal plan dimensions of 20 ft by 7 ft. Its 20-ft dimension will extend east–west along the pond edge, corresponding to the combined width of the two 8-ft staging areas and the central 4-ft starting zone. The 7-ft dimension will extend landward from the pond edge.

The area will be bounded by a low wooden retaining barrier, constructed provisionally from nominal 2-by-4 lumber or an equivalent material. The barrier will be oriented with its wider dimension vertical, providing a nominal retaining height of approximately 4 in.

The waterside portion of this barrier will run immediately along the concrete pond edge. The barrier will continue along the eastern and western sides of the containment area and along its landward boundary, forming a continuous enclosure.

The purpose of this outer barrier is to prevent golf balls that are accidentally dropped or otherwise deposited on the concrete walkway from rolling into the pond or escaping the designated containment area.

Transfer-Bot Access Corridor

A nominally 3-ft-deep access corridor will extend east–west along the pond edge within the concrete containment area.

This corridor will provide the working space needed for the transfer bot to approach the floating-ball and submerged-ball staging areas and transport recovered golf balls toward the designated dry-dock scoring pen.

The corridor will remain unobstructed by permanent field structures. Golf balls may nevertheless accumulate within the corridor during a heat, including those accidentally dropped by the transfer bot.

The transfer bot may move along this corridor as necessary to service both staging areas, subject to the robotic system requirements specified in Section 3.

Dry-Dock Scoring Pen

A separate 4-ft by 4-ft dry-dock scoring pen will be located within the outer concrete containment area.

The pen will be centered directly behind the team's 4-ft by 4-ft starting zone, with its eastern and western boundaries aligned with those of the starting zone.

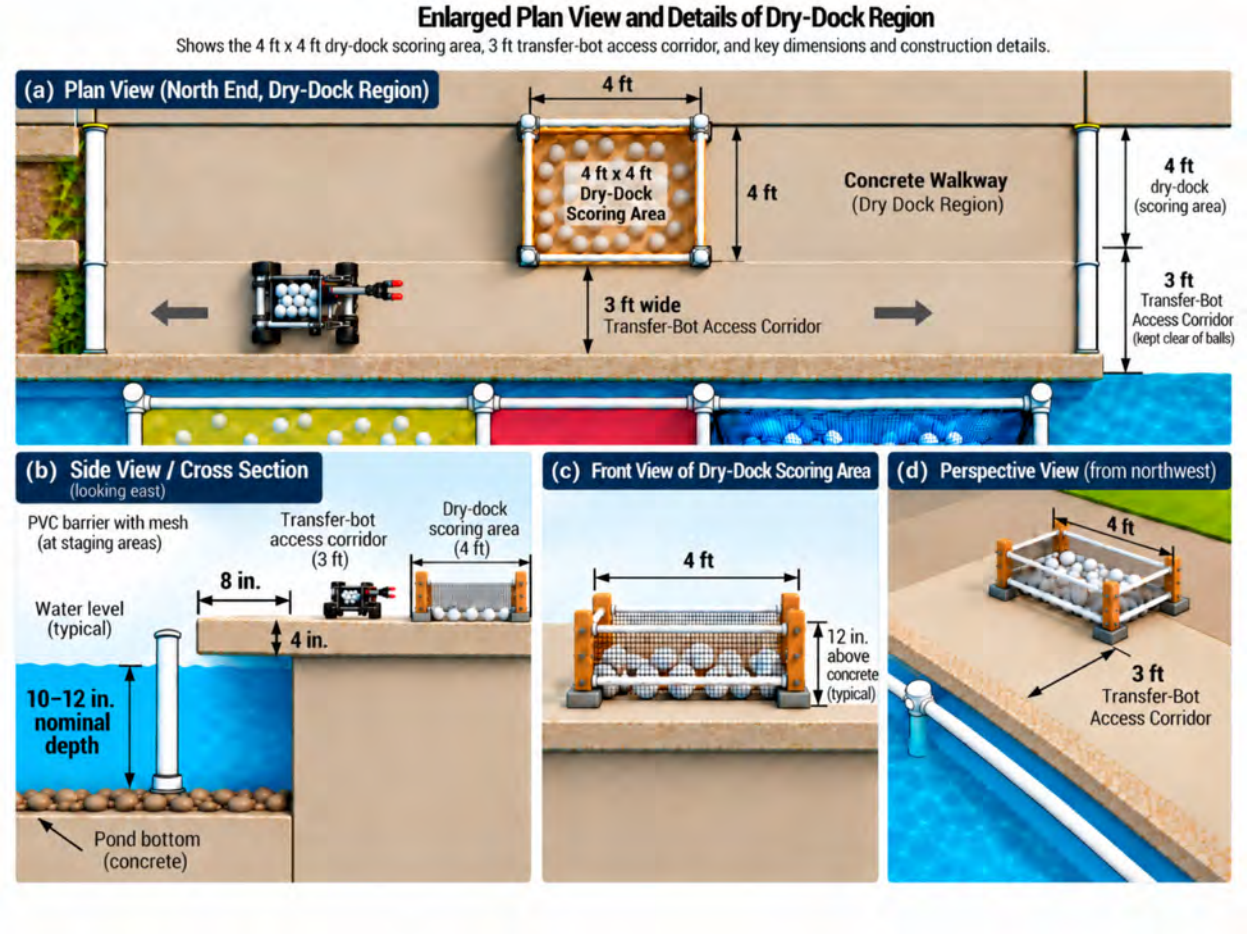


Figure 10: **Dry-dock layout and construction details.** (a) Enlarged plan view showing the 4-ft by 4-ft scoring pen and the 3-ft-deep transfer-bot access corridor. (b) Cross-section illustrating the concrete pond ledge and its approximately 8-in. horizontal overhang. (c) Front view of the scoring-pen enclosure, approximately 12 in. high. (d) Perspective view of the scoring pen. Dimensions and construction details are approximate.

The pond-facing edge of the scoring pen will be positioned approximately 3 ft landward of the concrete pond edge, leaving the transfer-bot access corridor between the pond and the scoring pen.

The scoring pen will be bounded by a raised, semi-rigid containment structure, constructed provisionally from PVC rails, vertical supports, and lightweight mesh. The enclosure will have a nominal height of approximately 12 in. and will be designed to retain golf balls deposited into the pen.

The interior of the scoring pen will remain open to accommodate the accumulation of both floating and submerged golf balls during a heat.

Scoring Regions

The designated concrete containment area and the dry-dock scoring pen will constitute two distinct scoring regions.

Golf balls resting within the 4-ft by 4-ft dry-dock scoring pen at the conclusion of a heat will receive the full dry-dock scoring value.

Golf balls resting elsewhere within the designated 20-ft by 7-ft concrete containment area, including the transfer-bot access corridor, will receive a reduced scoring value.

Golf balls that leave the designated containment area and come to rest on other portions of the concrete walkway will not receive dry-dock scoring credit.

Each golf ball will be scored according to its final location at the conclusion of the heat. Detailed scoring values and procedures are specified in Section 5.

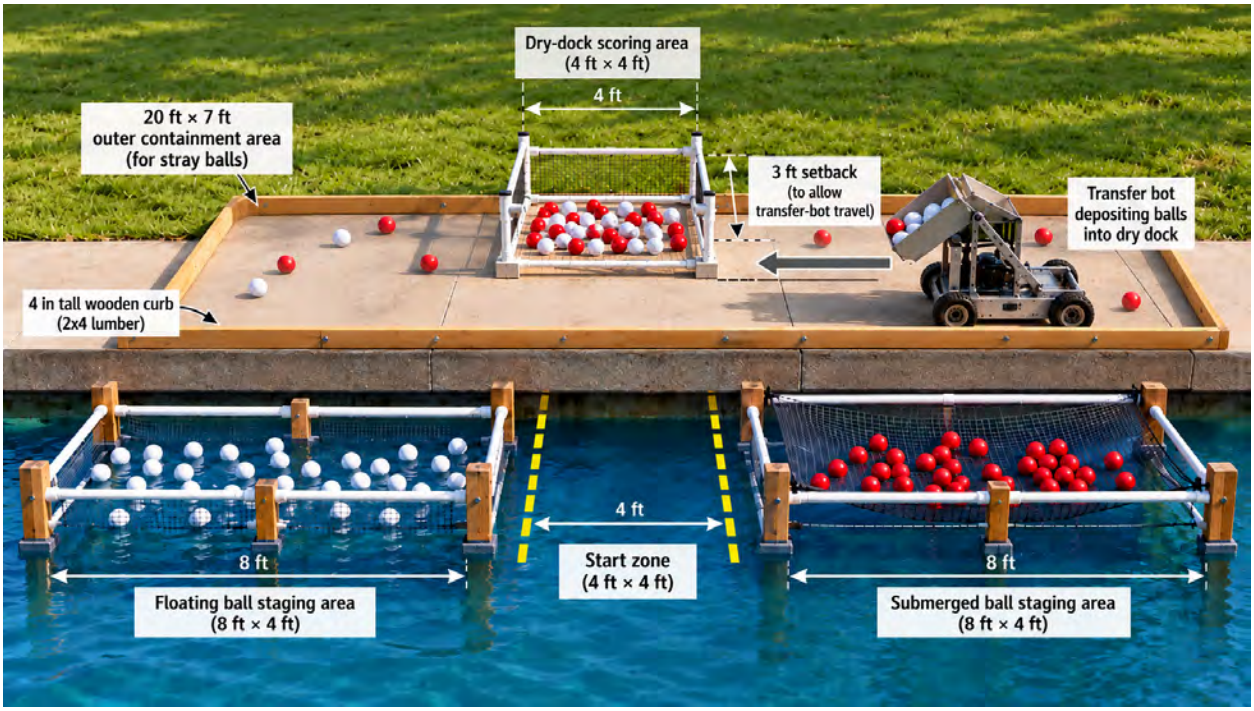


Figure 11: **Dry-dock containment and transfer-bot operating area.** Perspective illustration of the 20-ft by 7-ft concrete containment area bounded by a low wooden retaining curb. The 4-ft by 4-ft dry-dock scoring pen is positioned behind the 3-ft-deep transfer-bot access corridor. Golf balls deposited within the scoring pen receive full dry-dock credit, while balls elsewhere within the designated outer containment area receive reduced credit. The transfer bot and construction details are illustrative.

3 Robotic System Requirements

3.1 Three-Robot Requirement

Each team must develop three robotic systems: two collection bots and one dedicated transfer bot. The collection bots will operate within the pond, recovering floating and submerged golf balls and delivering them to the team's designated staging areas.

The transfer bot will be responsible for completing the final pond-to-land transfer of recovered golf balls to the designated dry-dock scoring region.

Teams may specialize their two collection bots for different tasks, such as floating-ball collection and submerged-ball dredging, or may develop bots with overlapping capabilities. The distribution of collection functions between the two bots is left to the team.

3.2 Collection Bots

The two collection bots will operate within the pond and will be responsible for recovering golf balls from the active playing field and delivering them to their team's designated ball recovery staging areas.

Each collection bot must fit within a maximum planform footprint of 16 in. by 24 in. at the start of each heat. The orientation of this footprint is unrestricted.

Collection bots may incorporate mechanisms for surface collection, dredging, storage, transport, and unloading as required by the team's design. Mechanisms may extend beyond the initial footprint after the heat begins.

The main body of each collection bot, including its hull or flotation structure, must remain outside the designated floating-ball containment enclosure. However, collection bots may extend ball-collection, handling, and transfer mechanisms into the designated staging areas as necessary.

Collection bots may not deposit golf balls directly onto the adjacent concrete walkway. The final pond-to-land transfer is reserved for the dedicated transfer bot.

3.3 Harbor Access and Operating Boundaries

Each team's harbor is reserved for the robotic systems assigned to that team. Collection bots may enter and leave their own harbor through its designated 4-ft-wide entrance.

Collection bots may operate throughout the shared central playing region but may not deliberately enter the opposing team's harbor. The harbor boundary is defined by the harbor-divider barrier and its entrance, including the full region extending from the barrier toward the corresponding north or south pond boundary.

Incidental contact between robots or field structures will be addressed separately under the

general competition rules. Deliberate interference with opposing robots or staging structures is prohibited.

3.4 Transfer Bot

The third robotic system will serve as the dedicated transfer bot and will be responsible for transferring recovered golf balls from the team's staging areas to its designated dry-dock scoring region.

A land-based transfer bot may operate within the designated concrete containment area, including the 3-ft-deep transfer-bot access corridor, and may incorporate east–west mobility as necessary to service both staging areas.

Alternatively, a team may position the transfer bot in the water within the rear portion of the designated starting zone, adjacent to the pond ledge. A transfer bot configured in this manner must remain stationary throughout the heat.

The transfer bot must fit within a maximum planform footprint of 24 in. by 48 in. at the start of each heat. If positioned within the starting-zone region, the 48-in. dimension must be oriented approximately east–west along the pond edge.

The transfer bot may extend ball-handling, lifting, and transfer mechanisms into the designated staging areas as necessary. However, it may not travel into the open playing field to collect golf balls directly from the pond.

The design of the transfer bot, including its mobility, reach, lifting, storage, and unloading mechanisms, is left to the team, subject to the physical operating boundaries and scoring requirements established elsewhere in these rules.

4 Match Procedure

4.1 Heat Format

Each competition heat will involve two teams operating simultaneously from opposite ends of the playing field. One team will be assigned to the north harbor and the other to the south harbor. Each heat will have a duration of 4 min 30 s.

During a heat, each team may operate its two collection bots to recover floating and submerged golf balls from the shared central playing region and return the recovered balls to its designated harbor.

The dedicated transfer bot may operate during the heat to move recovered golf balls from the staging areas onto the adjacent concrete containment area, including the designated dry-dock scoring pen. All collection and transfer activities must be completed within the 4 min 30 s competition period.

Each golf ball will be scored according to its final location or qualifying possession status

when the heat concludes, as specified in Section 5.

4.2 Starting Conditions

At the beginning of each heat, each team's two collection bots must be positioned entirely within its designated 4-ft by 4-ft starting zone, located inside the team's harbor. The dedicated transfer bot must also be positioned within its permitted starting and operating region, as specified in Section 3.

A total of 200 golf balls will be placed within the approximately 42-ft by 10-ft initial ball-distribution corridor centered in the shared playing region. These will consist of 100 white floating golf balls and 100 orange submerged golf balls.

The floating golf balls will initially be confined within this corridor by temporary floating barriers. The submerged golf balls will be distributed across the pebble-covered pond bottom within the same designated region.

At the starting signal, collection bots may depart their respective harbors through the designated 4-ft-wide entrances. Course staff will remove the temporary floating barriers as quickly and gently as practical. The barriers will be cleared before collection bots reach the central ball-distribution region.

The permanent harbor-divider barriers will remain in place throughout the heat. These barriers are intended to limit the migration of golf balls into the team harbors while preserving the designated entrances for collection-bot traffic.

4.3 Determination of Heat Winners

The team earning the greater number of points during a heat will be declared the winner. Scores will be determined according to Section 5, including any applicable partial-recovery credit.

If the two teams finish with identical scores, the winner will be determined using the following criteria, applied in order:

1. Greater number of submerged golf balls occupying valid scoring locations or qualifying for partial-recovery credit at the conclusion of the heat.
2. Greater number of submerged golf balls within the designated concrete containment area, including the dry-dock scoring pen, at the conclusion of the heat.
3. Greater number of submerged golf balls within the designated dry-dock scoring pen at the conclusion of the heat.

If the heat remains tied after application of these criteria, an additional tiebreaking heat will be conducted under the standard competition rules. This procedure applies to both round-robin and elimination heats, ensuring that every heat produces a winner.

4.4 Tournament Format

The competition will consist of four teams. The tournament will begin with a round-robin phase in which each team will compete against every other team once, resulting in six heats.

The purpose of the round-robin phase is to establish team rankings and determine seeding for the subsequent single-elimination championship tournament. All four teams will advance to the elimination phase.

Teams will be ranked first according to the number of round-robin heats won. If two or more teams finish with the same number of wins, their relative rankings will be determined using the following tiebreaking criteria, applied in order:

1. Greater total number of points scored across all round-robin heats.
2. Greater cumulative number of submerged golf balls occupying valid scoring locations or qualifying for partial-recovery credit at the conclusion of round-robin heats.
3. Greater cumulative number of submerged golf balls within the designated concrete containment areas, including the dry-dock scoring pens, at the conclusion of round-robin heats.
4. Greater cumulative number of submerged golf balls within the designated dry-dock scoring pens at the conclusion of round-robin heats.

If a tie remains after application of these criteria, an additional seeding procedure will be established by the course staff.

The four teams will then enter a single-elimination championship tournament. The semifinal pairings will be determined by the round-robin rankings:

- Semifinal 1: First seed versus fourth seed.
- Semifinal 2: Second seed versus third seed.

The winners of the two semifinal heats will advance to the championship final. The winner of the final heat will be declared the ME72 Dredge Challenge competition champion.

4.5 Timeouts

During round-robin play, each team may call a maximum of two timeouts. Each timeout will delay the start of the team's scheduled heat by 5 min.

At the beginning of the single-elimination championship tournament, each team will receive a new allotment of two 5-min timeouts for use during elimination play.

Timeouts must be called before the start of a heat. Once a heat has begun, the competition clock will not be stopped except under circumstances specified by the course staff.

5 Scoring

5.1 Golf-Ball Scoring

Scoring will be based on the final location or possession status of each golf ball at the conclusion of the 4 min 30 s heat. Each ball may receive credit only once. Golf balls that remain in the general playing field, or are otherwise outside a designated scoring region and not retained within a robot, will receive no points.

Three principal scoring levels will be recognized:

1. **Ball Recovery Staging Areas.** Golf balls recovered to the appropriate in-water staging area within a team's harbor will receive intermediate-recovery credit.
2. **Outer Concrete Containment Area.** Golf balls transferred onto the team's designated concrete containment area, but remaining outside the dry-dock scoring pen, will receive increased scoring credit.
3. **Dry-Dock Scoring Pen.** Golf balls successfully deposited within the designated 4-ft by 4-ft dry-dock scoring pen will receive the highest scoring credit.

In addition, limited partial-recovery credit will be awarded for golf balls retained within a robot when the heat expires, even if they have not reached a designated scoring region.

Submerged golf balls will carry twice the scoring value of floating golf balls at every level, including partial-recovery credit. The scoring values are summarized in the following table.

Table 1: Revision 1 golf-ball scoring rubric.

Golf Ball Type	Retained in Robot	Recovery Staging Area	Concrete Area Outside Pen	Dry-Dock Scoring Pen
Floating golf ball	0.1	1.0	1.5	2.0
Submerged golf ball	0.2	2.0	3.0	4.0

Recovery Staging Areas

A floating golf ball will receive intermediate-recovery credit only if it remains within the team's designated floating-ball containment enclosure at the conclusion of the heat.

A submerged golf ball will receive intermediate-recovery credit only if it remains within the designated submerged-ball receiving net at the conclusion of the heat. Golf balls resting on the pond bottom beneath or adjacent to the receiving net will not receive scoring credit.

A floating golf ball that enters and remains within the designated containment enclosure will receive credit regardless of whether it arrived through deliberate collection, wave action, or incidental contact.

Concrete Containment Area

Golf balls resting within the designated 20-ft by 7-ft concrete containment area, but outside the dry-dock scoring pen, will receive the intermediate concrete-area scoring value shown in the table.

This region includes the 3-ft-deep transfer-bot access corridor and other portions of the designated concrete containment area outside the scoring pen.

Golf balls that leave the designated concrete containment area and come to rest elsewhere on the surrounding walkway will receive no points.

Dry-Dock Scoring Pen

Golf balls resting within the designated 4-ft by 4-ft dry-dock scoring pen at the conclusion of the heat will receive the full dry-dock scoring value shown in the table.

A golf ball must remain within the scoring pen when the heat ends to qualify for this credit.

Partial-Recovery Credit

Golf balls retained within either of a team's collection bots or its dedicated transfer bot at the conclusion of a heat will receive partial-recovery credit.

Each floating (white) golf ball retained within a robot will receive 0.1 point. Each submerged (orange) golf ball retained within a robot will receive 0.2 point.

To qualify, a golf ball must be physically retained within the robot's collection, conveyance, or storage mechanism when the heat expires. A ball merely touching a robot, being pushed through the water, or passing through an open, non-retaining guide or funnel will not qualify.

Partial-recovery credit recognizes successful ball acquisition without completed delivery to a designated scoring region. This credit represents 10% of the corresponding staging-area value for either ball type.

Robots must cease operation at the conclusion of the heat and retain their remaining payloads until course staff have completed the scoring inspection.

Final Scoring

Each golf ball will be counted only once, according to its final scoring location or qualifying possession status at the conclusion of the heat. Points will not be accumulated for successive stages of recovery.

The total team score will equal the sum of the individual point values of all golf balls occupying valid scoring locations or qualifying for partial-recovery credit when the heat expires.

The determination of heat winners, round-robin rankings, and championship tournament seeding is governed by the procedures specified in Section 4.

To qualify for a designated scoring region, a golf ball must be entirely within that region at the conclusion of the heat. Balls resting on or across the boundary of a scoring region will be evaluated by course staff according to this requirement.

6 General Rules and Safety

6.1 Robot Interaction and Interference

The competition is intended to encourage active robotic operation within a shared playing field. Collection bots from opposing teams may encounter one another while navigating, collecting golf balls, and returning to their respective harbors.

Incidental collisions and brief physical contact between opposing collection bots are permitted. Robots may push against one another while competing for the same floating or submerged golf balls, or while maneuvering for position in the shared playing region. Such contact will not constitute a rules violation provided that it occurs during normal collection or navigation activities.

Deliberate attempts to damage, overturn, disable, entrap, or otherwise incapacitate an opposing robot are prohibited. Intentional ramming, sustained pushing unrelated to normal collection activities, grappling, or using field structures to pin an opposing robot are also prohibited.

Each team's harbor is reserved for its own robotic systems. Collection bots may enter and leave their assigned harbor through the designated 4-ft-wide entrance but may not deliberately enter the opposing team's harbor.

Deliberate obstruction of an opposing team's harbor entrance is prohibited. Robots may travel through or temporarily occupy the water immediately outside an opposing harbor during normal collection and navigation activities. However, a robot may not intentionally position itself or extend its mechanisms across the entrance for the purpose of preventing opposing collection bots from entering or leaving.

Incidental collisions and brief competitive contact may occur near a harbor entrance during normal collection and navigation. Such contact is permitted, including brief pushing between robots competing for nearby golf balls. Proximity to a harbor entrance does not, by itself, constitute a violation. Deliberate or sustained interference intended to prevent a robot from entering or leaving its harbor remains prohibited.

Robots must not intentionally interfere with an opposing team's ball recovery staging areas, transfer bot, or dry-dock scoring structures.

6.2 Field Structures and Harbor Protection

The competition field includes permanent floating boundaries, harbor-divider barriers, ball recovery staging structures, and dry-dock containment and scoring structures. Teams must design and operate their robots with appropriate consideration for these elements.

Incidental contact with field structures during normal robot operation is permitted. However, robots may not deliberately strike, displace, lift, detach, damage, or otherwise interfere with field structures.

Collection bots must use the designated 4-ft-wide entrance when entering or leaving their assigned harbor. Robots may not deliberately cross over, pass underneath, or otherwise bypass a harbor-divider barrier.

The main body of each collection bot, including its hull or flotation structure, must remain outside the floating-ball containment enclosure, as specified in Section 3. Collection and transfer mechanisms may extend into the staging areas during normal ball-handling operations.

Robots may not intentionally remove golf balls from an opposing team's staging areas, concrete containment area, or dry-dock scoring pen.

6.3 Human Intervention

Once a heat has begun, team members may not physically touch, reposition, retrieve, repair, or otherwise assist their robots unless authorized by course staff.

If a robot becomes disabled, stranded, or entangled with another robot or a field structure, the team may request assistance from course staff. Any intervention will be performed at the discretion of course staff, with safety taking precedence over continued competition.

Course staff may intervene to address unsafe conditions, restore displaced field structures, retrieve disabled robots, or resolve other conditions affecting the conduct of a heat.

Golf balls may occasionally migrate into a team's harbor through its designated entrance. Such balls may remain within the harbor during the heat and will not receive scoring credit unless they reach a designated scoring location or qualify for partial-recovery credit.

Course staff may retrieve or redistribute stray golf balls when necessary for safety or orderly competition. Such intervention will be performed only when it can be accomplished safely and without providing an unfair advantage to either team.

Teams may not manually collect, move, or deposit competition golf balls while a heat is in progress.

6.4 Electrical and Battery Safety

All robots must be designed and constructed for safe operation in the competition environment, including exposure to water, wet surfaces, and nearby spectators.

Electrical connections, motor controllers, battery terminals, and other vulnerable electrical components must be appropriately protected against water exposure.

Each robot must incorporate a clearly identifiable and readily accessible means of disabling propulsion and powered mechanisms. Teams must be able to shut down their robots promptly when instructed by course staff.

Batteries must be securely mounted and protected against mechanical damage, short circuits, and unintended water contact. Damaged, swollen, leaking, or otherwise unsafe batteries may not be used.

Battery charging must be performed in designated areas and in accordance with applicable manufacturer guidance and course safety requirements.

Exposed electrical conductors and unprotected battery terminals are prohibited.

6.5 Inspection and Rule Interpretation

All robotic systems will be subject to inspection by course staff before competition. Inspection may include verification of robot footprint dimensions, mechanical construction, electrical safety, shutdown capability, and compliance with competition operating restrictions.

Teams must demonstrate that their robots can be operated and disabled safely. Course staff may require corrective action before permitting a robot to participate in a heat.

At the conclusion of each heat, all robots must cease motion and powered mechanism operation. Teams must preserve golf balls retained within their robots until scoring inspection has been completed, as required by Section 5.

Course staff will determine each team's final score according to the scoring provisions established in Section 5.

Violations of competition rules may result in warnings, point deductions, disqualification from a heat, or other corrective measures determined by course staff according to the nature and severity of the violation.

Course staff will seek to apply scoring, inspection, and competition rules consistently to all teams. Any corrective actions or penalties will take into account the circumstances of the incident, including whether the violation was incidental, deliberate, or repeated.

The course instructor will serve as the final authority on interpretation of the competition rules and the resolution of disputes.